

**1.10**

Rational Functions and Holes

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

| Teacher: Mohammad Abu-

Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A canceled factor records an excluded input in the original function.

2. Write one precise sentence explaining this principle: The hole y-coordinate comes from the simplified formula evaluated at the excluded input.

3. Write one precise sentence explaining this principle: Equivalent expressions need not define equal functions if their domains differ.

4. Write one precise sentence explaining this principle: A removable discontinuity can be “filled” by defining the missing value appropriately.

Core

5. Find the hole of $\frac{x^2-9}{x-3}$.
6. Define $F(2)$ to make $F(x) = \frac{x^2-4}{x-2}$ continuous.
7. Can a rational function have a hole and a vertical asymptote at the same x-value?
8. State what the formula $R(x) = \frac{(x-a)P(x)}{(x-a)Q(x)}$, $x \neq a$ tells you and name any required restriction.
9. State what the formula hole at $\left(a, \frac{P(a)}{Q(a)}\right)$ if $Q(a) \neq 0$ tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Rational Functions and Holes. State its domain and justify one key feature.

Medium

11. Find every hole and any remaining vertical asymptote of $\frac{x^2-16}{x-4}$.
12. Find every hole and any remaining vertical asymptote of $\frac{(x+2)(x-5)}{x+2}$.
13. Find every hole and any remaining vertical asymptote of $\frac{x^2+x-6}{x-2}$.
14. Find every hole and any remaining vertical asymptote of $\frac{(2x-1)(x+3)}{x+3}$.

15. Find every hole and any remaining vertical asymptote of $\frac{(x-1)^2(x+4)}{(x-1)(x+2)}$.

Hard

16. Define $F(3)$ to remove the discontinuity in $F(x) = \frac{x^2-9}{x-3}$.

17. Find k so $\frac{x^2+kx-12}{x-3}$ has a hole at $x=3$.

18. Explain why $\frac{(x-2)^2}{x-2}$ and $x-2$ have the same graph except at one point but are not equal as functions.

19. A graph follows $y=x^2+1$ except for a hole at $x=-2$. Write a rational formula.

Difficult

20. Find all k for which $\frac{x^2+(k-2)x-2k}{x-k}$ has a hole at $x=k$.

21. Construct a rational function with holes at $x=-1$ and $x=2$, with hole y-values 3 and -3.

22. Can a hole occur at a point where the simplified formula has a vertical asymptote? Explain.

Challenge / HOT

23. Prove that redefining a rational function at finitely many holes does not change its end behavior.

24. Find a rational expression that simplifies to $x + 1$ but has exactly three holes at prescribed inputs a, b, c .

AP-Style Multiple Choice

25. The hole of $(x^2 - 1)/(x - 1)$ is

- (A) $(-1, 0)$
 - (B) $(1, 2)$
 - (C) $(1, 0)$
 - (D) none
26. A hole results from
- (A) noncanceled denominator zero
 - (B) fully canceled common factor
 - (C) numerator degree
 - (D) horizontal asymptote
27. To find hole y-coordinate, use
- (A) original undefined fraction
 - (B) simplified formula
 - (C) denominator only
 - (D) leading coefficients
28. Equivalent expressions define equal functions only if
- (A) degrees match
 - (B) domains also match
 - (C) graphs have same end behavior
 - (D) coefficients are integers
29. $(x - 2)/(x - 2)^2$ has at $x=2$
- (A) a hole
 - (B) a vertical asymptote
 - (C) an intercept
 - (D) a finite value
30. A removable discontinuity can be removed by
- (A) changing all outputs
 - (B) assigning the missing limit value
 - (C) adding an asymptote
 - (D) deleting the numerator

AP-Style Free Response

- 31.** Let $R(x) = \frac{(x-1)(x+2)^2}{(x-1)^2(x+2)}$. (a) Simplify with restrictions. (b) Classify $x=1$ and $x=-2$. (c) State one-sided behavior or hole coordinate.

32. A function equals $\frac{x^2+ax+b}{x-2}$ for x not equal 2 and is defined by $F(2)=7$. Find a and b so F is continuous at 2 and has zero at $x=-3$.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.